

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company discovers that nearly 70% of defects originate before coding begins. Which software engineering activity should receive the highest priority to reduce future defects?

- A. Deployment
- B. Requirements and Design Review
- C. Code Optimization
- D. User Training

Answer: B

2. According to software engineering principles, why is software often called a “logical product”?

- A. It is stored in databases.
- B. It has no physical form.
- C. It requires electricity.
- D. It is executed by processors.

Answer: B

3. Which Software Development Process Model is MOST suitable when developers expect customer requirements to evolve throughout the project?

- A. Waterfall
- B. Agile
- C. V-Model
- D. Linear Sequential

Answer: B

4. In Rapid Application Development (RAD), components are primarily developed:

- A. Sequentially
- B. Randomly
- C. In Parallel
- D. After Deployment

Answer: C

5. Which of the following is the BEST example of a technical requirement?

- A. Increase annual revenue.
- B. Reduce customer complaints.
- C. Integrate with a REST API.
- D. Improve employee satisfaction.

Answer: C

6. During Requirements Engineering, which phase produces documents such as SRS or PRD?

- A. Validation
- B. Specification
- C. Negotiation
- D. Elaboration

Answer: B

7. Which Requirements Engineering challenge is MOST likely when developers have little understanding of the client's business domain?

- A. Scope Baseline
- B. Lack of Domain Knowledge
- C. Continuous Delivery
- D. Configuration Management

Answer: B

8. Which statement BEST describes an Agile iteration?

- A. It contains only coding.
- B. It delivers a potentially usable product increment.
- C. It ends with project deployment.
- D. It removes customer involvement.

Answer: B

9. In Agile development, a timebox mainly prevents:

- A. Customer feedback
- B. Unlimited expansion of iteration duration
- C. Software testing
- D. Code refactoring

Answer: B

10. In XP, postponed improvement ideas are formally documented during which phase?

- A. Planning
- B. Exploration
- C. Productionizing
- D. Death

Answer: C

11. Which XP practice requires programmers to immediately merge completed code into the shared codebase?

- A. Coding Standards
- B. Continuous Integration
- C. Metaphor
- D. Pair Programming

Answer: B

12. Which Scrum artifact remains unchanged until the current Sprint finishes?

- A. Product Vision
- B. Product Backlog
- C. Sprint Backlog
- D. Release Plan

Answer: C

13. Which Scrum phase includes defining architecture before Sprint development begins?

- A. Game Phase
- B. Sprint Review
- C. Pre-game Phase
- D. Post-game Phase

Answer: C

14. Which document is produced after reviewing a functional prototype in DSDM?

- A. Test Log
- B. Functional Prototyping Review Document
- C. Coding Standard
- D. Product Backlog

Answer: B

15. Which DSDM activity develops a DESIGN PROTOTYPE suitable for end-user testing?

- A. Business Study
- B. Functional Model Iteration
- C. Design and Build Iteration
- D. Feasibility Study

Answer: C

16. In FDD, the overall project schedule is primarily constructed based on:

- A. Source Code Size
- B. Individual Features
- C. Database Tables
- D. Customer Budget

Answer: B

17. Which FDD role develops internal tools that improve developer productivity?

- A. Domain Expert
- B. Tester
- C. Toolsmith
- D. Chief Architect

Answer: C

18. Which software myth incorrectly assumes that purchasing the latest development tools guarantees project success?

- A. Customer Myth
- B. Professional Myth

- C. Management Myth
- D. Maintenance Myth

Answer: C

19. According to software engineering studies, why are defects found during deployment generally more expensive to fix?

- A. Hardware becomes obsolete.
- B. More users are affected and changes propagate throughout the system.
- C. Programming languages become outdated.
- D. Source code cannot be modified.

Answer: B

20. Which Requirements Engineering practice establishes an officially approved set of requirements before development proceeds?

- A. Story Mapping
- B. Requirements Baselineing
- C. Sprint Planning
- D. Pair Programming

Answer: B